

Finite-dimensional algebra in the unital singleton case

`MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_algebra_of_singleton_alongNonUnital`

theorem

Distinguishes finite dimension of the representation space from finite dimension of the algebra embedded in its operators.

Statement

Let A be a nonzero unital complex C^* -algebra, let H be a separable complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ satisfy the singleton condition of R11. Then A is finite dimensional over $\mathbb{C}$.

Assumptions

Finite dimensionality of A or H is not an initial hypothesis. Neither faithfulness nor separability of A is assumed.

Conclusion

The vector space A itself has a finite complex basis.

Proof route

First obtain finite dimensionality of H . The compact-operator conclusion gives injectivity of π , so A embeds linearly into the finite-dimensional space $\mathcal{B}(H)$.

▼ Proof steps

1. Apply The all-representations singleton condition implies the unital singleton condition to the given condition. For a unital nonzero irreducible $*$ -representation ρ , use the same operator map $\rho_0(a) = \rho(a)$ through the interface that does not require a unit equation. It is still nonzero irreducible, so the hypothesis supplies the required unitary equivalence. Thus the same π satisfies the unital singleton condition.
2. Apply Finite dimension of the unital singleton representation space with this singleton π , nonzero ordered unital A and separable H to obtain a finite basis of H . Hence the space $\mathcal{B}(H)$ of bounded complex-linear operators is finite dimensional: if $\dim_{\mathbb{C}} H = n$, its dimension is n^2 .
3. The theorem Faithfulness of a unital singleton model gives injectivity of the same complex-linear π . Thus

$$A \xhookrightarrow{\pi} \mathcal{B}(H), \quad \dim_{\mathbb{C}} \mathcal{B}(H) < \infty.$$

An injective linear map into a finite-dimensional space forces its domain to be finite dimensional. This is the final step of Finite dimension of the faithfully represented algebra, which the exact theorem applies after Step 1.

Main citations

- Exact theorem and proof
- The all-representations singleton condition implies the unital singleton condition
- Finite dimension of the unital singleton representation space
- Faithfulness of a unital singleton model

Lean source signature (exact)

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theorem finiteDimensional_algebra_of_singleton_amongNonUnital
  [Nontrivial A] [TopologicalSpace.SeparableSpace H]
  (pi : Representation A H)
  (hsingle : IsSingletonIrreducibleModelAmongNonUnital.{u, v, u} pi) :
  FiniteDimensional ℂ A
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[TopologicalSpace.SeparableSpace H]	The complete complex Hilbert representation space H is separable.
(pi : Representation A H)	The specified unital $*$ -representation $\pi : A \rightarrow B(H)$.
(hsingle : IsSingletonIrreducibleModelAmongNonUnital.{u, v, u} pi)	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
FiniteDimensional ℂ A	The conclusion is finite dimension of the algebra A , as opposed to the intermediate finite dimension of H .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

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